

Supplementary Material for “Statistical Certification of Autonomous Vehicle Fallback Triggers: Closed-Loop Evaluation Under Dataset Shift”

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This document contains the exploratory and post hoc analyses summarized in Section V of the main paper. All numbers use the same settings as the main paper ($\alpha = 0.05$, $\delta = 0.10$, 1,994 Argoverse 2 calibration and 1,996 test scenarios, 1,504 nuScenes scenarios) and can be regenerated from the released per-scenario records without re-simulation.

S1. CLASS-CONDITIONAL CERTIFICATION (POST HOC)

Table S1 lists the differences between the closest prior work and this study.

TABLE S1
LUO *et al.* VERSUS THIS STUDY.

	Luo <i>et al.</i>	This paper
Guarantee	$\Pr[\text{no alert} \mid \text{unsafe}] \leq \epsilon$, class-conditional	$\Pr[\text{harm not pre-empted}] \leq \alpha$ over scenarios, high probability
Unsafe label	safety score of logged future $< f_0$	nuPlan harm in the no-fallback run
Predictor	Trajectron++	AutoBot
Alert consequence	none (scored on logs)	MRM executed in closed loop
Cost of alerting	false-positive rate	unnecessary stops, induced collisions, residual harm
Data	nuScenes, Lyft	Argoverse 2, nuScenes, Waymo
Shift	label shift only	dataset shift, weighted repair
Latency	not studied	studied

Table S2 implements the certificate of Luo *et al.* (cited in the main paper) in our framework: Mondrian split conformal prediction on the unsafe class, where unsafe means harm in the reference run and an alert means firing before it, with the finite-sample correction and without their tie randomization. At $\epsilon = 0.05$ on the 316 unsafe calibration scenarios it selects a threshold equivalent to always firing, so the class-conditional target is vacuous for this predictor at this level. The score separates harmful from harmless scenarios well enough for a marginal target (AUROC 0.83) but not for a 5% miss rate among harmful ones. At the marginal-matched $\epsilon = \alpha/\pi = 0.32$ it selects the same threshold as empirical tuning. Deployed on nuScenes, that certificate has a conditional miss rate of 66% against its 32% target. The harm base rate rises from 15.8% to 22.1%, but the conditional miss rate is far above what that alone explains, so the class-conditional guarantee does not survive the dataset shift either.

S2. REVERSE SHIFT (EXPLORATORY, POST HOC)

We also certified on half of the nuScenes scenarios and deployed on Argoverse 2 test, a direction pre-registered as

TABLE S2
MARGINAL CERTIFICATE (LTT) VERSUS CLASS-CONDITIONAL CERTIFICATES (LUO-STYLE), THRESHOLDS CERTIFIED ON ARGOVERSE 2. MISS: MARGINAL MISS RATE; COND.: MISS RATE AMONG SCENARIOS WITH HARM IN THE REFERENCE RUN. RATES IN %.

Certificate	Argoverse 2 test			nuScenes		
	Miss	Cond.	Unnec.	Miss	Cond.	Unnec.
LTT, marginal $\alpha = 0.05$	3.7	24.2	28.5	13.2	58.1	26.1
Luo-style, $\epsilon = 0.05$	0.0	0.0	84.9	5.0	19.6	77.9
Luo-style, $\epsilon = 0.32$	4.3	28.5	26.0	14.9	66.0	21.3

a reverse-shift analysis. As pre-registered, with all scenarios, the procedure *refused to certify* on both halves, because of the 5.0% floor. On the avoidable subset (post hoc), thresholds certified on nuScenes ($n \approx 715$) give an Argoverse 2 miss rate of 1.9% and 2.0%, both below the target, at 40.6% and 39.5% unnecessary stops. A certificate from the harder dataset therefore transfers safely but conservatively, while the reverse does not transfer. This asymmetry is a single-pair observation pending replication.

S3. RECALIBRATION WITH A FEW LABELED TARGET SCENES

Table S3 draws k labeled target scenes at random and evaluates on the rest. The nuScenes analysis (avoidable subset, 60 draws) was exploratory and not pre-registered. Its Waymo counterpart (all scenarios, 200 draws) was pre-registered as R1–R4 before any Waymo outcome was read, and its outcomes are reported in Section V of the main paper. On nuScenes, uncertified tuning on 40–60 target scenes exceeds the target in about two draws in three. A certified threshold on the same scenes keeps the target (violations 0–10%) but stops in 53–81% of scenarios, against 27% without target labels. A handful of labeled scenes therefore restores a *valid* certificate, not the original operating point. On Waymo, uncertified tuning violates the target in 69.5% and 59% of draws with 40 and 60 target scenes, and weighting the Argoverse 2 calibration set with 60 Waymo scenes ($n_{\text{eff}} = 232$) keeps validity at 0% violation.

S4. SIMULATOR SENSITIVITY

Three changes to the simulator, each paired by scenario seed against the main run (reactive traffic, 4.0 m/s², 2 Hz), leave every conclusion of the main paper unchanged. Non-reactive log-replay traffic raises the harm base rate (17.7%

TABLE S3

RECALIBRATION WITH $k = 60$ LABELED TARGET SCENES. nuScenes: EXPLORATORY, AVOIDABLE SUBSET, 60 DRAWS. WAYMO: PRE-REGISTERED (R1–R4), ALL SCENARIOS, 200 DRAWS. VIOL.: FRACTION OF DRAWS WITH TEST MISS $> \alpha$. RATES IN %.

Method	nuScenes			Waymo		
	Miss	Viol.	Unnec.	Miss	Viol.	Unnec.
AV2-certified, no labels	9.2	–	27.1	2.7	0	45.0
Target-only tuned	6.0	70	36.7	6.2	59	29.6
Target-only LTT	0.8	0	72.5	1.5	2.5	58.6
AV2 + 60, weighted LTT	2.8	0	53.3	1.7	0	49.8

versus the main run’s 15.1 %, since logged agents cannot react to an ego that deviates from its log) and, with it, unnecessary stops by 7.9 points [95 % CI 5.3, 10.8]. The miss rate itself does not move (paired difference -0.5 points $[-1.3, 0.0]$). A 2.0 m/s^2 comfort deceleration changes neither miss nor stops (paired difference exactly 0 in both), because both alert-level quantities depend only on the reference run and the firing tick (Observation 1 of the main paper), not on how hard the vehicle then brakes. It does lower induced collisions (2.6 % to 1.6 %), the one quantity that does depend on post-fire kinematics. Decisions at 10 Hz on 100 scenarios give a miss rate of 2.9 % [1.0, 8.2], consistent with the main run within an interval widened by the smaller sample.

S5. CERTIFYING RESIDUAL HARM DIRECTLY (POST HOC)

The certificate in the main paper bounds the alert indicator $L(X, \lambda)$, not harm in the run the vehicle then drives. We therefore tried to certify *residual harm* $H(X, \lambda)$, the indicator that the nuPlan harm definition fires anywhere in the executed run, before or after the fallback engages. H is not monotone in λ , because firing earlier can itself create harm (4.5 % of fresh calibration scenarios have a non-monotone H), so Conformal Risk Control does not apply by construction. Learn-then-Test needs no monotonicity but is tested on the same fresh Argoverse 2 split as the replication analysis of the main paper ($n_{\text{cal}} = n_{\text{test}} = 904$, target $\alpha_H = 0.05$, $\delta = 0.10$). The comparisons were pre-registered as N1–N5 before running.

TABLE S4

CERTIFYING RESIDUAL HARM ON THE FRESH SPLIT. RATES IN %, WILSON 95 % INTERVALS. VIOL.: FRACTION OF 200 RESPLITS WHOSE TEST RESIDUAL HARM EXCEEDS 5 %. THE SECOND PREDICTOR GIVES THE SAME PICTURE (OUTCOME-LEVEL LTT: 2.8 % HARM, 84.8 % STOPS, 16.0 % VIOLATIONS).

Certificate on	Resid. harm [CI]	Miss	Stops	Viol.
Alert L , LTT (main paper)	5.0 [3.7, 6.6]	3.4	29.0	–
Residual harm H , LTT	2.8 [1.9, 4.1]	–	84.8	15.0
Residual harm H , CRC	3.9 [2.8, 5.3]	–	37.5	45.5

Three results, each contrary to what we pre-registered, follow from Table S4. First, certifying H at 5 % with LTT selects a threshold at which the vehicle fires in essentially every scenario (84.8 % unnecessary stops), and even then it exceeds the target in 15 % of resplits, above δ , so it is not a valid certificate (N1 refuted). Second, the alert-level threshold’s residual harm on this fresh split is 5.0 % [3.7, 6.6], so its interval includes the 5 % target, unlike the 6.5 %

on the main split (N2 refuted on fresh data; the interval is wide enough that a modest excess is not excluded). Third, CRC violates in 45.5 % of resplits, but LTT also exceeds δ , so a non-monotone-aware calibrator is not by itself sufficient (N3 refuted). A joint certificate on residual harm and induced collisions over threshold and two MRM decelerations (2 and 4 m/s^2) found no configuration other than always firing (N4 refuted). The practical reading is that, with this score and this maneuver, residual harm of 5 % or less is not reachable by tuning the trigger threshold alone at a useful operating point. A different maneuver or score is needed, not a different calibrator.

S6. SHIFT SEVERITY AND THE COST OF THE INTERVENTION (DESCRIPTIVE)

These two tables are computed from the stored records and test no hypothesis.

Shift severity. Table S5 measures how far each target is from the Argoverse 2 calibration set in the seven pre-deployment covariates used for the density ratio. The domain classifier is evaluated by 5-fold cross-validated AUROC (0.5 means indistinguishable), and the Argoverse 2 test split is a null control. nuScenes is much easier to tell apart than Waymo, has a Kish effective sample size of 66 against 288, and a 5.0 % floor of harm that no trigger can pre-empt. The certificate failed on the more distant target and held on the nearer one, which is consistent with shift severity mattering, but two targets cannot establish that, and the floor alone accounts for 5.0 of the 13.2 points missed on nuScenes.

TABLE S5

SHIFT SEVERITY AND THE ARGOVERSE 2-CERTIFIED THRESHOLD ($\lambda = 1.47$). AUROC: CROSS-VALIDATED DOMAIN CLASSIFIER AGAINST THE ARGOVERSE 2 CALIBRATION SET. FLOOR: HARM PRESENT EVEN WHEN THE FALLBACK FIRES AT THE FIRST DECISION TICK. RATES IN %.

Target	n	AUROC	n_{eff}	Harm	Floor	Miss [CI]
AV2 test (control)	1,996	0.50	1,978	15.1	0.0	3.7 [2.9, 4.6]
Waymo	1,000	0.91	288	15.4	0.1	2.7 [1.9, 3.9]
nuScenes	1,504	0.97	66	22.1	5.0	13.2 [11.6, 15.0]

Scenarios from the same scene are not independent. The Wilson intervals in the main paper treat scenarios as independent. The nuScenes scenario identifiers contain the scene name, which allows a check: the 1,504 nuScenes scenarios in the main analysis come from only 37 scenes (41 per scene on average). Resampling whole scenes (cluster bootstrap, 10,000 draws) widens the interval for the miss rate of the Argoverse 2-certified threshold from [11.6, 15.0] to [6.1, 21.0], which still excludes the 5 % target, but only just. A second, largely disjoint set of nuScenes scenarios (1,342 scenarios from 23 scenes, one scene in common, collected for the replication analyses) gives a miss rate of 9.5 % with Wilson interval [8.1, 11.2] and cluster-bootstrap interval [3.9, 16.5], which includes the target. The first-tick floor also differs between the two samples (5.0 % against 0.0 %), so it varies strongly with which scenes are drawn. The direction of the failure therefore replicates, but with a few dozen scenes per sample the size of the nuScenes failure is far less certain than the main-paper intervals suggest. Argoverse 2 and Waymo identifiers

are opaque, so the same check, and group-level resplits for the validity analysis, were not possible there.

Where the induced collisions come from. At the certified threshold on Argoverse 2 test, 2.6 % of scenarios [1.9, 3.3] suffer an induced collision. All 129 induced contact events are other vehicles striking the stopped ego, 99 from behind and 30 from the side. They concentrate where the ego is slow or fires early. Scenarios in the two lower terciles of initial ego speed (the lowest starts at standstill, the middle one below 4.6 m/s) have induced rates of 3.6 % [2.4, 5.3] and 3.5 % [2.3, 5.1], against 0.6 % [0.2, 1.5] in the upper tercile, and scenarios in which the fallback fires at or before the median firing tick (tick 5, 2.5 s) have 9.2 % [6.8, 12.4] against 3.0 % [1.7, 5.1] for later firing. These contacts do not look like a follower caught off guard by hard braking. Of the 129, only 3.9 % occur within 2 s of the fallback firing, the median is 8.3 s after it, and 78 % come more than 5 s after it, by which time a vehicle braking at 4 m/s² from city speed has long been stationary. The 129 contacts come from only 51 scenarios, and 67 % of them are in scenarios where the ego starts at standstill, so the fallback there changes little about what the ego does. We therefore cannot attribute this cost to the act of stopping. It may instead come from simulated traffic, driven along logged paths with a car-following model, that does not yield to a vehicle which stays stationary in its path for many seconds, which a real driver would also be expected to avoid (released with the code). The 2.6 % induced rate should be read as an upper bound on the cost that stopping itself causes in this simulator. The fallback also costs progress: where it fires, the vehicle stops for 7.4 s on average and completes 49 % of the route, against 94 % where it does not. Collision severity (closing speed) was not recorded in the stored rollouts, so we cannot grade these collisions, and the rates count scenarios, not contacts.

S7. DIRECT EXECUTION AGAINST THE LOOKUP

The main paper evaluates every trigger by lookup in a table of forced-fire rollouts, which is exact when a run under the trigger coincides with the reference run until it fires (Observation 1). The assumption is measured in the forced runs but was not tested by executing a trigger. This check, pre-registered as Amendment 7 before any run, does so. It draws 61 test scenarios at random from the 2,000 stored Argoverse 2 test rows (the job was stopped at 60 complete scenarios for compute reasons, independently of any outcome), and runs four geometric triggers live with the predictor scoring at every decision tick and no score replay or forced-fire table: the certified LTT threshold with zero and with one decision tick of latency, and the thresholds at the 25th and 75th percentile of the calibration peak scores (about 75 % and 25 % of scenarios fire). Each run is scored with the same nuPlan harm definition and compared, scenario by scenario, with the lookup at the same threshold and latency. No direct run failed.

Table S6 shows no disagreement in any alert-level quantity or in the induced collisions. The aggregate rates are identical for every trigger (for example 6.6 % missed and 18.0 % unnecessary stops for the certified threshold, 8.2 % missed with one tick of latency). One pair differs in harm anywhere in the

TABLE S6
DIRECT EXECUTION VERSUS LOOKUP, 61 SCENARIOS $\times$ 4 TRIGGERS = 244 PAIRS. DISAGREEMENT: PAIRS IN WHICH THE TWO DIFFER.

Quantity	Disagree	Wilson 95 % upper bound
Firing step (or no firing)	0 / 244	1.5 %
Missed intervention	0 / 244	1.5 %
Unnecessary stop	0 / 244	1.5 %
Induced collision	0 / 244	1.5 %
Harm anywhere in the run	1 / 244	2.3 %

run (seed 1934, certified threshold): the lookup records harm after the fallback and the direct run does not. We did not trace its cause, but a rare pre-fire difference of this kind is what the drift measurement in Section III-E of the main paper reports (the pre-fire time-to-collision trace differed in 1.2 % of forced runs). Route completion is identical in 120 of the 244 pairs and differs by 0.0001 on average and 0.018 at most. The pre-registered rule, that the lookup is refuted as a substitute if any quantity disagrees in more than 5 % of pairs, is therefore not triggered. The sample is small, with a 95 % upper bound on the disagreement rate of about 1.5 % for the alert-level quantities, covers one dataset, one predictor and one maneuver, and does not test stochastic traffic, for which assumption (A1) would not hold and the lookup would estimate the outcome under a different random draw than the one the trigger would have met. The magnitudes of the pre-fire deviations in the forced runs were not stored, only whether they exceeded 1 cm.

S8. REPLICATION ON FRESH DATA AND A SECOND PREDICTOR

Table S7 gives the numbers behind the replication analysis in the main paper. Violation frequencies over 200 resplits have Wilson 95 % intervals of [4.6, 12.0] for LTT with the primary predictor and [5.8, 13.8] with the second, so, as in the main split, the point estimates meet $\delta = 0.10$ but their Monte Carlo intervals do not exclude a violation frequency above it. The uncertified baselines (46–51.5 %) are far outside it. The gap between the geometric and the confidence trigger also replicates: with the same target, LTT stops 29.0 % [26.1, 32.0] of scenarios against 67.0 % [63.9, 70.0] for the tuned confidence trigger on the fresh split, and 30.6 % [27.7, 33.7] against 68.3 % [65.1, 71.2] with the second predictor.

TABLE S7
REPLICATION ON FRESH, DISJOINT DATA ($n_{\text{cal}} = n_{\text{test}} = 904$).
VIOLATION: FRACTION OF 200 RESPLITS WITH TEST MISS $> \alpha$.

Method	Fresh held-out		2nd predictor	
	Miss (%)	Viol. (%)	Miss (%)	Viol. (%)
LTT (geometric)	3.4	7.5	3.9	9.0
CRC (geometric)	4.3	46.0	4.6	47.5
Tuned geometric	4.3	51.5	4.6	51.0
T1 confidence	4.8	50.0	4.6	46.0

S9. CERTIFYING UNDER THE DEPLOYED LATENCY

The main paper shows that a certificate calibrated at zero latency fails when the actuation chain is slower. Because the loss at latency ℓ decision ticks is the stored miss indicator at

tick $\tau(X) + \ell$, LTT can be run directly on that loss without new simulation (Amendment 8, pre-registered before the run). We certify on the Argoverse 2 calibration set at $\ell \in \{0, 1, 2\}$ ticks (0, 0.5, 1.0 s), deploy at the same latency on the test set, and repeat over the same 200 half-and-half resplits as the main paper.

TABLE S8

LTT CALIBRATED AT THE DEPLOYED LATENCY (“AWARE”) VERSUS THE ZERO-LATENCY CERTIFICATE DEPLOYED AT LATENCY ℓ (“NAIVE”), ARGOVERSE 2 TEST. MISS AND STOPS ON THE TEST SET, RATES IN %. VIOL.: FRACTION OF 200 RESPLITS WITH TEST MISS ABOVE 5 %, WITH WILSON 95 % INTERVALS.

ℓ	Naive miss	Naive viol.	Aware miss	Aware stops	Aware viol.
0	3.7	9.5 [6.2, 14.4]	3.7	28.5	9.5 [6.2, 14.4]
1	5.3	92.0 [87.4, 95.0]	3.6	34.4	12.0 [8.2, 17.2]
2	9.1	100 [98.1, 100]	4.1	42.5	11.5 [7.8, 16.7]

The zero-latency certificate is not valid under latency, as claimed. The latency-aware certificate brings the miss rate back below 5 % on the test set, at a cost of 6 and 14 points of unnecessary stops. Its violation frequency, however, is 12.0 % and 11.5 % at one and two ticks, above the $\delta = 0.10$ we pre-registered, so by our own rule the claim that latency-aware certification restores validity is refuted. Three readings are possible and we cannot separate them. The Wilson intervals include 0.10. The statistic counts resplits whose finite test half exceeds 5 %, not resplits whose population risk does, and the zero-latency case is already 9.5 %. And the loss at a positive latency is less smooth in λ , which may weaken the fixed-sequence test in practice. The guarantee itself is unchanged, since the loss is still a fixed function of the scenario and the threshold, but we report the frequency as measured. Across resplits the mean unnecessary-stop rate of the latency-aware certificate is 27.6, 31.8 and 57.0 %, so at two ticks the cost depends strongly on the calibration draw.